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Detecting Marine pollutants and Sea Surface features with Deep learning in Sentinel-2 imagery

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ABSTRACT

Despite the significant negative impact of marine pollution on the ecosystem and humans, its automated detection and tracking from the broadly available satellite data is still a major challenge. In particular, most research and development efforts focus on one specific pollutant implementing, in most cases, binary classification tasks, e.g., detect *Plastics* or no *Plastics*, or target a limited number of classes, such as *Oil Spill*, *Look-alikes* and *Water*. Moreover, most developed algorithms tend to operate successfully only locally, failing to scale and generalize adequately towards operational deployments. Our aim is to address these challenges by introducing a holistic approach towards marine pollutant detection using remote sensing. We argue that constructing such operational solutions requires detectors trained and tested against different types of pollutants, various sea surface features and water-related thematic classes. We offer such a Marine Debris and Oil Spill (MADOS) dataset, composed of high-resolution multispectral Sentinel-2 (S2) data, consisting of 174 scenes captured between 2015 and 2022, with approximately 1.5 M annotated pixels, which are globally distributed and collected under various weather conditions. Moreover, we propose a novel Deep Learning (DL) framework named MariNeXt, based on recent state-of-the-art architectural advancements for semantic segmentation, which outperforms all baselines by at least 12 % in F_1 and mIoU metrics. The extensive quantitative and qualitative validation justifies our choices and demonstrates the high potential of the proposed approach. We further discuss the underlying discrimination challenges among the competing thematic classes. Our dataset, code and trained models are openly available at https://marine-pollution.github.io/.

1. Introduction

Marine pollutants, such as oil and litter, have been a global concern due to their adverse impact on marine life, human health and the economy (Barboza et al., 2018; Beaumont et al., 2019; Shahidul Islam and Tanaka, 2004). This global threat has been recognized in the UN Sustainable Development Goals (SDGs) under Goal 14, i.e., to conserve and use the oceans, seas and marine resources for sustainable development (Walker, 2021). In particular, the main objective by 2025 is the prevention and significant reduction of marine pollution of all kinds.

Effective marine monitoring systems are crucial for tackling marine pollution and achieving sustainable goals. Detecting and monitoring marine pollution have already received a vast amount of research and development efforts. Current approaches rely mainly on in situ campaigns via conventional methods, e.g., ships, nets, trawls, laboratory analysis etc. (Cadiou et al., 2020; Eriksen et al., 2018; Parinos et al., 2019) as well as remote sensing data due to several and significant benefits. Indeed, satellite remote sensing provides cost-effective and frequent measurements of the sea surface at large spatial scales, supporting successfully automated detection systems (Krestenitis et al., 2019; Sannigrahi et al., 2022; Temitope Yekeen and Balogun, 2020; Wang and Hu, 2021a).

However, most remote sensing methods have been designed to detect a single sea surface feature or a small number of categories without taking into consideration other competing classes. This fact usually results in several false positives e.g., natural *Water* formations are classified as *Plastics*, *Water* class is misclassified as *Other* class (Heiselberg, 2020; Sannigrahi et al., 2022; Solé Gómez et al., 2022). Operational marine monitoring solutions should be designed to accurately identify any sea surface feature or water-related class while being false-alarm

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resistant under complex weather and sea state conditions. Additionally, in some cases, marine classes[1] present similar spectral and spatial patterns which pose additional challenges on classification tasks in optical satellite images. For instance, Sea snot has been recently considered to significantly influence Marine Debris detection (Hu et al., 2022). At the same time, the discrimination of floating matters in radar data is regarded by far more difficult (Qi et al., 2022).

Towards exploring the unique characteristics of marine pollutants and other floating materials, spectral analysis has been performed with optical imagery, also considering the competing thematic classes (Hu, 2022; Qi et al., 2020; Qi and Hu, 2021). Apart from the spectral methods, only a limited number of studies have considered advanced Machine Learning (ML) techniques that exploit other patterns (e.g., spatial) and try to address the detection of challenging cases such as small and texture-complex floating objects (Kikaki et al., 2022; Mifdal et al., 2021; Mikeli et al., 2022).

Moreover, a small number of optical multispectral datasets exist that are openly available to support spectral analysis studies or research towards marine monitoring solutions (Kikaki et al., 2022; Liu et al., 2017; Mifdal et al., 2021). This is because identifying and annotating sea surface features in optical satellite data is quite challenging due to the properties of current sensors (e.g., S2 different band resolutions) (Hu, 2022, 2021), the complex optical properties of specific features (Dierssen, 2019; Garaba and Dierssen, 2018), and the rare events of specific floating objects (e.g., large plastic patches).

To this end, we aim to confront the aforementioned challenges by introducing a new benchmark dataset with several competing marine classes that formulates a challenging *semantic segmentation* task using sparse annotations. We also propose a novel DL framework for the detection of the considered thematic classes. Specifically, we make the following contributions:

- We introduce a new well-curated benchmark dataset with 15 diverse thematic classes: e.g., floating *Marine Debris*, *Oil Spill*, *Sargasssum* macroalgae, *Ship*, *Oil Platform*, *Sea snot*, water-related classes, etc., named MADOS, which is globally distributed, based on 174 Sentinel-2 scenes collected under different weather and sea conditions, consisting of approximately 1.5 M annotated pixels.
- We propose a novel deep detection framework called MariNeXt based on the SegNeXt (Guo et al., 2022) architecture, which incorporates three improvement strategies in modeling, training, and testing, outperforming current ML baselines by 12 % and delivering quite promising mapping outcomes.
- We elaborate and discuss important findings regarding the discrimination potentials of the considered thematic classes based on extensive quantitative and qualitative evaluation.

2. Background

In the following subsections, we briefly refer to existing marine pollution datasets (section 2.1) and methods (section 2.2) focusing both on pollutants (i.e., *Oil Spill*, *Marine Debris*) and on sea surface features (i.e., *Sargassum* macroalgae, *Sea snot*, *Ships Waves*, *Wakes*) detection on high-resolution S2 imagery.

2.1. Marine pollutants datasets

Tables 1 and 2 summarize recent efforts addressing *Oil Spill* and *Marine Debris* detection with multispectral S2 data. In particular, they demonstrate that most studies were based on a relatively limited number of S2 images. For *Marine Debris,* (Arias et al., 2021; Booth et al., 2023; Duarte and Azevedo, 2023; Kikaki et al., 2020, 2022; Mifdal et al., 2021; Mikeli et al., 2022; Rußwurm et al., 2023; Sannigrahi et al., 2022; Solé Gómez et al., 2022) employed more than 15 S2 scenes, while for *Oil Spill* only Sun et al. (2018) utilized a similar amount of S2 scenes (Tables 1 and 2). Additionally, most studies were limited to a single up to a three-year period, while Kikaki et al., (2020,2022); Sun et al. (2018) collected *Marine Debris* and *Oil Spill* data for more than six years. Finally, for most studies, the annotated/ reference data were sparse and restricted to one or a few small geographical regions (Tables 1 and 2). We highlight that most of the considered works do not provide publicly available data. Specifically, no Sentinel-2 open-access dataset exists aiming to benchmark *Oil Spill* detection methods. For *Marine Debris*, Kikaki et al. (2022) introduced MARIDA, an open-access benchmark dataset that discriminates marine litter from several competing classes at a global scale. Mifdal et al. (2021) also openly provided annotations of floating objects (one general class) using globally S2 imagery.

2.2. Detection of Marine pollutants and other sea surface features

Oil Spill. *Oil spill* detection from satellite imagery has been extensively studied over the past few decades (Al-Ruzouq et al., 2020). Specifically, radar sensors dominate due to their all-weather and all-day capabilities (Dong et al., 2022). Nevertheless, employing optical data is crucial to separate oil from look-alikes and extract information about oil types and thickness (Garcia-Pineda et al., 2020; Zhao et al., 2014). This subsection reviews recent studies for *Oil spill* detection based on S2 optical imagery presented in Table 1.

To detect *Oil Spill*, most methods have been focused on spectral indices (Kolokoussis and Karathanassi, 2018; Rajendran et al., 2021a; Tysiąc et al., 2022) and spectral bands ratios (Abou Samra and Ali, 2022; Rajendran et al., 2021c, 2021b) (Table 1). Support Vector Machine (SVM), Gradient Boosting Algorithm (Argamosa et al., 2022; Tysiąc et al., 2022) and Parallelepiped Supervised Algorithm (PSA) (Rajendran et al., 2021c) have also been applied for *Oil Spill* detection. Most studies were designed to detect one *Oil Spill* class while only a few (Rajendran et al., 2021a; Tysiąc et al., 2022) considered five (i.e., *Oil Spill*, *Water*, *Snow*, *Vegetation*, *Wetlands)* or six (i.e., *Oil Spill*, *Water*, *Ground*, *Vegetation*, *Buildings*, *Snow*) classes.

Marine Debris. Compared to *Oil Spill*, detecting *Marine Debris* from spaceborne data is a more recent research field, with S2 MSI being the most frequently used satellite sensor for monitoring marine litter in coastal waters (Mukonza and Chiang, 2022). In particular, many efforts have been concentrated on artificial targets, coupled with various methods e.g., Spectral Indices (SI) (Themistocleous et al., 2020), Spectral Unmixing (SU), Matched Filtering (MF) (Topouzelis et al., 2020), and ML (Basu et al., 2021; Jamali and Mahdianpari, 2021). In a different direction, other studies have focused on detecting floating Marine Debris in real cases using S2 images (Table 2).

Regarding ML, well-established supervised learning techniques, i.e., Naïve Bayes, SVM, Random Forest (RF), U-Net, DeeplabV3+, as well as Generative Adversarial Networks (GANs) have been applied for various *Marine Debris* detection tasks (Table 2). Additionally, recently, (Booth et al., 2023; Mikeli et al., 2022; Olyaei et al., 2022; Rußwurm et al., 2023) proposed novel ML solutions boosting the performance on the open-access MARIDA dataset (Kikaki et al., 2022).Concerning the number of classes, Biermann et al. (2020); Ciappa (2021); Gonzaga et al. (2021) considered five classes, while Kikaki et al. (2022); Mikeli et al. (2022) targeted 11 different thematic classes.

Other sea surface features. To extract *Sargassum* from S2 imagery and remove noise patterns, a fully automated workflow has been suggested using Convolutional Neural Networks (CNNs) (Wang and Hu, 2021a). Furthermore, RF has been applied to distinguish *Sargassum* from *Ulva* (Xiao et al., 2022), and VGGUnet has been employed for *Sargassum* extraction in high-resolution optical data (Wang and Hu, 2021b). Reflectance spectra of *Sea snot/ mucilage* were examined by Hu et al. (2022) while spectral indices (Kavzoglu and Goral, 2022), OBIA (Kavzoğlu et al., 2021) and SVM (Tuzcu Kokal et al., 2022) were adopted for *Sea snot* detection.

[1] The term 'marine classes' refers to all classes considered in our analysis.

Table 1
Publications for *Oil Spill* (Oil) mapping using Sentinel-2 data during the period 2018–2022.

	General Info				Pre-processing		Classification/ Detection	
Publication	Main purpose	Study Area	Time span	# of S2 scenes	Atm. Corr.	Reflectance	Method	# of classes
Kolokoussis & Karathanassi, 2018	Oil detection	Zakynthos Isl. & Saronikos Gulf (Greece)	2017	4	No	TOA	OBIA, SI	1
Sun et al., 2018	Oil detection	Gulf of Mexico	2004–2016	**24**	Yes	Rayleigh	Statistical analysis	1
Althawadi & Hashim, 2019	Oil detection & Calibration	Arabian Gulf	2016–2017	7	Yes	Surface Refl.	Region-based segmentation, Regression	1
Rajendran et al., 2021a	Oil detection	Ambarnaya river, Russia	2020	14	No	TOA	SI, Dynamical permafrost model, DSM	5
Rajendran et al., 2021c	Oil detection	Mauritius	2020	6	Yes	Surface Refl.	PSA, Band ratios, DSM	3
Rajendran et al., 2021b	Oil detection	Mauritius	2020	8	No	TOA	Band ratios, DSM	1
Argamosa et al., 2022	Oil detection	Manila Bay, Philippines	2020	2	No	TOA	PCA, Gradient boosting algor.	2
Abou Samra and Ali, 2022	Oil detection	Nile Delta	2019–2021	N/A	Yes	Rayleigh	Band ratios, DSM	1
Tysiąc et al., 2022	Oil detection	Ambarnaya river, Russia	2020	6	No	TOA	SVM, SI	**6**

Abbreviations: OBIA = Object-Based Image Analysis, SI = Spectral Indices, DSM = Decorrelation Stretch Method, PSA = Parallelepiped Supervised Algorithm, PCA = Principal Component Analysis, N/A = Not Mentioned, SVM = Support Vector Machine.

For *Ship* identification, DL techniques on S2 data have already been assessed (Štepec et al., 2019; Xie et al., 2020). CNNs have been examined to distinguish ships from icebergs (Heiselberg, 2020), and self-supervised learning was exploited to generate binary ship/no-ship segmentation outputs (Ciocarlan and Stoian, 2021). Other competing classes, such as *Waves* and *Wakes,* have been studied based on SVMs (Heiselberg and Heiselberg, 2017; Song et al., 2014), while Liu et al. (2021) proposed a Radon transform method for *Wake* identification, effectively eliminating false alarms such as *Waves* and *Oil Spill*.

3. Dataset and methods

3.1. The MADOS benchmark dataset

3.1.1. Data processing & curation

Satellite sensors with high spatial, spectral and temporal resolution are required to develop operational solutions for monitoring the marine environment. Currently, the openly available Copernicus S2 multispectral data cover global coastal waters at high spatial resolutions (10 m, 20 m and 60 m). Additionally, S2 satellite has a frequent revisit time of approximately five days (depending on the latitude), delivering a large amount of free geospatial data. It also acquires data in 13 spectral bands, which range from the visible to the Short-Wave Infrared (SWIR). Therefore, in our study, we utilized the S2 multispectral satellite images. We collected S2 Level 1C products (Top-of-Atmosphere reflectance) through EarthExplorer (USGS). Similar to previous studies on floating materials detection (Qi et al., 2020; Sun et al., 2018; Zhao et al., 2014), we applied ACOLITE atmospheric processor (Vanhellemont and Ruddick, 2018) to retrieve the Rayleigh reflectance.

Data annotation, curation and refining were performed following Kikaki et al. (2020) & (2022) methods and protocols. The annotation was performed by three image-interpretation experts with access to all collected data. To enhance the accuracy and reliability of our dataset, with a particular focus on *Marine Debris* and *Oil Spill* classes, we employed an inter-annotators agreement protocol. Specifically, for *Marine Debris* and *Oil Spill*, the intersection per two annotators was extracted (i.e., an agreement between at least two annotators regarding the class label). This approach enriched fine-grained annotations of *Marine Debris*, offering greater detail and precision. As *Oil Spill* was also a primary category in our focus on marine pollution, we applied this protocol to this class in a similar manner. We also used the same protocol to similarly appearing classes to *Marine Debris*, i.e., *Natural Organic Material* and *Sparse Floating Algae*. For the other classes, the union of the annotated data was calculated.

To improve the quality of our annotated data, the structure of the high-dimensional observations was visualized. Thus, we embedded our data in a 2D space, identifying the spectral patterns and outliers. Specifically, we utilized the t-distributed Stochastic Neighborhood Embedding (t-SNE) algorithm (van der Maaten and Hinton, 2008) using Spectral Angle Mapping (SAM) as a distance metric.

Based on the annotated data, we cropped the S2 scenes into non-overlapped image patches. In the produced dataset, the initial spatial resolution of each spectral band was retained without any upsampling technique. Each provided image corresponds to 2.4x2.4 km^2; thus, the 10 m spectral bands form image patches of 240x240 pixels, the 20 m patches of 120x120 pixels and the 60 m patches of 40x40 pixels.

3.1.2. MADOS thematic classes and annotations

We have constructed MADOS, which contains annotated S2 data based on 174 scenes and 47 tiles captured between 2015 and 2022. Numerous annotations were created consisting of 1,481,155 pixels based on studies, reports and data on marine pollution as well as intensive photo-interpretation (Table S1). Finally, our dataset is distributed over 22 countries (Fig. 1) and contains 15 different thematic classes (Table 3).

We focus on two major marine pollutants, i.e., *Oil Spill* and *Marine*

Table 2
Publications for *Marine Debris* (MD) mapping using Sentinel-2 data during the period 2018–2022.

General info					Pre-processing		Classification/ Detection	
Publication	Main purpose	Study Area	Time span	# of S2 scenes	Atm. Corr.	Reflectance	Method	# of classes
Kikaki et al., 2020	MD sources & pathways detection	Honduras Gulf	2014–2019	37	Yes	Surface Refl.	In situ data, Image interpretation, Spectral signatures	3
Biermann et al., 2020	MD detection	Global	2018–2019	12	Yes	Surface Refl.	Naïve Bayes	5
Mifdal et al., 2021	Float. Objects detection	Global	2018–2020	15	Yes	Surface Refl., TOA	SVM, RF, Naïve Bayes, U-Net	2
Gonzaga et al., 2021	MD detection	Manila Bay, Philippines	2019–2020	6	Yes	Surface Refl.	Naïve Bayes, MTMF, FDI, NDVI, PI	5
Arias et al., 2021	Windrows detection	Greece, Honduras	N/A	40	Yes	Surface Refl., TOA	Wasp Spectral Index	1
Ciappa, 2021	MD detection	Hawaii, N. Adriatic, S. Domingo	2020	6	Yes	Surface Refl.	Red edge bands, NDVI, PI, FDI	5
Kikaki et al., 2022	MD detection	Global	2015–2021	**63**	Yes	Rayleigh	RF, U-Net	**11**
Mikeli et al., 2022	MD detection	Global	2015–2021	**63**	Yes	Rayleigh	RF, U-Net	**11**
Olyaei et al., 2022	MD detection	Global	2015–2021	**63**	Yes	Rayleigh	Deep Knockoff, SVM, RF	3
Sannigrahi et al., 2022	MD detection	Mediterranean Sea	2018–2019 & 2021	27	Yes	Surface Refl.	SVM, RF	3
Solé Gómez et al., 2022	River debris detection	Drina, Los Angeles & Yangtzee	2018–2020	30	Yes	Surface Refl.	U-Net, U-Net3DE, DeepLab V3+	3
Ciappa, 2022	MD detection	N.Adriatic, Australia	2020	6	Yes	Surface Refl.	Spectral bands	4
Rußwurm et al., 2023	MD detection	Global	2015–2021	**89**	Yes	Surface Refl.	U-Net, Label Refinement Module	1
Booth et al., 2023	MD detection & density mapping	Global	2015–2021	**63**	Yes	Surface Refl.	U-Net, MAP-Mapper	1
Duarte & Azevedo, 2023	MD detection	Global	2016–2021	25	Yes	Surface Refl.	GANs, SI, XGBoost	7

Abbreviations: SVM = Support Vector Machine, RF = Random Forest, MTMF = Mixture Tuned Matched Filtering, FDI = Floating Debris Index, NDVI = Normalized difference vegetation index, PI = Plastic Index, GANs = Generative Adversarial Networks, SI = Spectral Indices, XGBoost = Extreme Gradient Boosting, N/A = Not Mentioned.

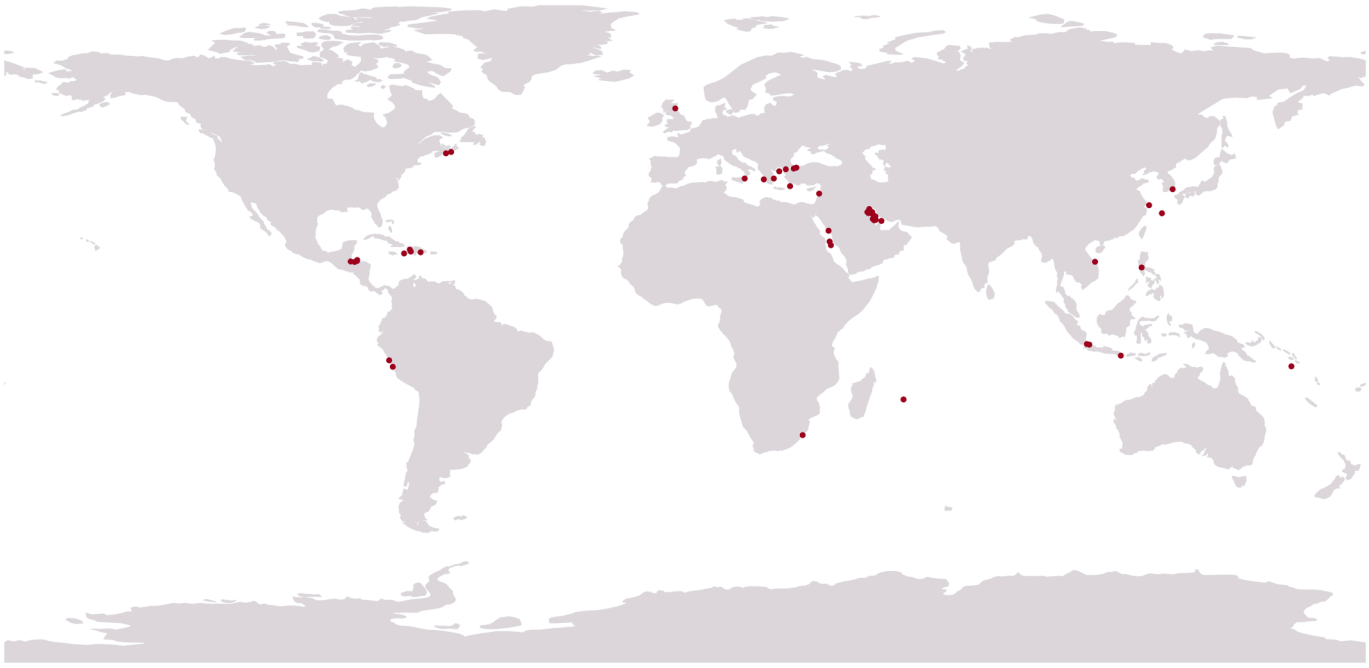


Fig. 1. The globally distributed Sentinel-2 tiles from which MADOS data were sampled.

Table 3
MADOS Thematic Classes. Per-class description, acronym, number of image patches and corresponding annotated pixels.

#	Class Name	Acronym	Description	# of patches (240x240)	# of pixels
1	Marine Debris	**MD**	Floating plastics, mixed anthropogenic debris	472	4696
2	Oil spill	**Oil**	Oil types including crude oil, emulsified, oil on water, metallic, sheen & rainbow	361	234,568
3	Dense *Sargassum*	**DenS**	Dense floating *Sargassum* macroalgae	77	4308
4	Sparse Floating Algae	**SpFA**	Sparse *Sargassum* & other macroalgae	157	4972
5	Natural Organic Material	**NatM**	Vegetation & Wood	92	1454
6	Ship	**Ship**	Sailing & Anchored Vessels	298	13,222
7	Marine Water	**MWater**	Clear Water & sea state Look-alikes	1360	496,702
8	Sediment-Laden Water	**SLWater**	High-Sediment river discharges with brown colour	57	376,061
9	Foam	**Foam**	Foam recorded at river fronts or coastal wave breaking area	58	1225
10	Turbid Water	**TWater**	Turbid waters close to coastal areas	232	170,306
11	Shallow Water	**SWater**	Coastal waters, including coral reefs and submerged vegetation	140	122,753
12	Waves & Wakes	**Waves**	Waves & Ship Wakes	224	20,374
13	Oil Platform	**OilPlat**	Offshore Oil Platform	348	15,701
14	Jellyfish	**Jellyfish**	Jellyfish	12	2541
15	Sea snot	**Sea snot**	Sea snot/ mucilage	63	12,272
	Total			2803	1,481,155

Debris. First, oil slicks from different sources (i.e., ship accidents, offshore platforms, discharges due to natural disasters and seafloor natural seepages) were recorded. Our aim was to develop a model capable of detecting one *Oil Spill* thematic class and not different oil types. However, oil forms of different thicknesses were considered and annotated in MADOS, including all categories established by NOAA (2016): a) Dark, b) Mousse (emulsified), c) Transitional dark, d) Metallic, e) Rainbow and f) Sheen (Fig. 2).

Regarding *Marine Debris*, we used the open-access MARIDA dataset (Kikaki et al., 2022), which distinguishes litter from various sea features. We also increased the number of *Marine Debris* pixels by adding new observations based on recent reports and publications (Table S1).

Following previous studies (Biermann et al., 2020; Hu et al., 2022; Kikaki et al., 2020; Martínez-Vicente et al., 2019), we additionally considered floating materials that have been suggested to be spectrally similar to plastics, or are occasionally aggregated with floating macroplastics on the ocean surface. Thus, *Foam*, *Sargassum*, *Natural Organic Material* (including driftwood), *Ships* and *Sea Snot* trajectories (Hu et al., 2022) were annotated (Fig. 2). Suspected *Jellyfish* investigated by Qi et al., (2020) were also taken into account to evaluate the discrimination from *Marine Debris* and *Sea Snot,* as previous studies have indicated the potential of spectral similarities (Hu et al., 2022; Van der Zande et al., 2014). Classes sometimes observed concurrently with oil slicks, such as *Oil Platform* and *Sparse Floating Algae,* were also considered and added. Moreover, sea state look-alikes were annotated and aggregated with the *Marine Water* class. Other water-related classes recorded on our study regions (i.e., *Sediment-Laden Water, Turbid Water, Shallow Water, Waves and Ship Wakes)* have also been annotated. Also, sparse *Sargassum* was grouped with other sparse floating algae species, forming a *Sparse Floating Algae (SpFa)* super-class. By adding these competing cases, we aim to enable the developed algorithms to discriminate marine pollutants from the other floating materials and sea surface features. We note that clouds and cloud shadows were not included in our analysis. We consider this a different, quite challenging task that the remote sensing scientific community is working on. Therefore, one could integrate any of the standard software tools (such as MAJA, Fmask, Sen2Cor) into the processing pipeline to address cloud and shadow detection.

MADOS consists of realistic observations recorded under diverse weather conditions aiming to support real-world applications (Fig. 2). However, due to this diversity as well as due to S2 specifications our data

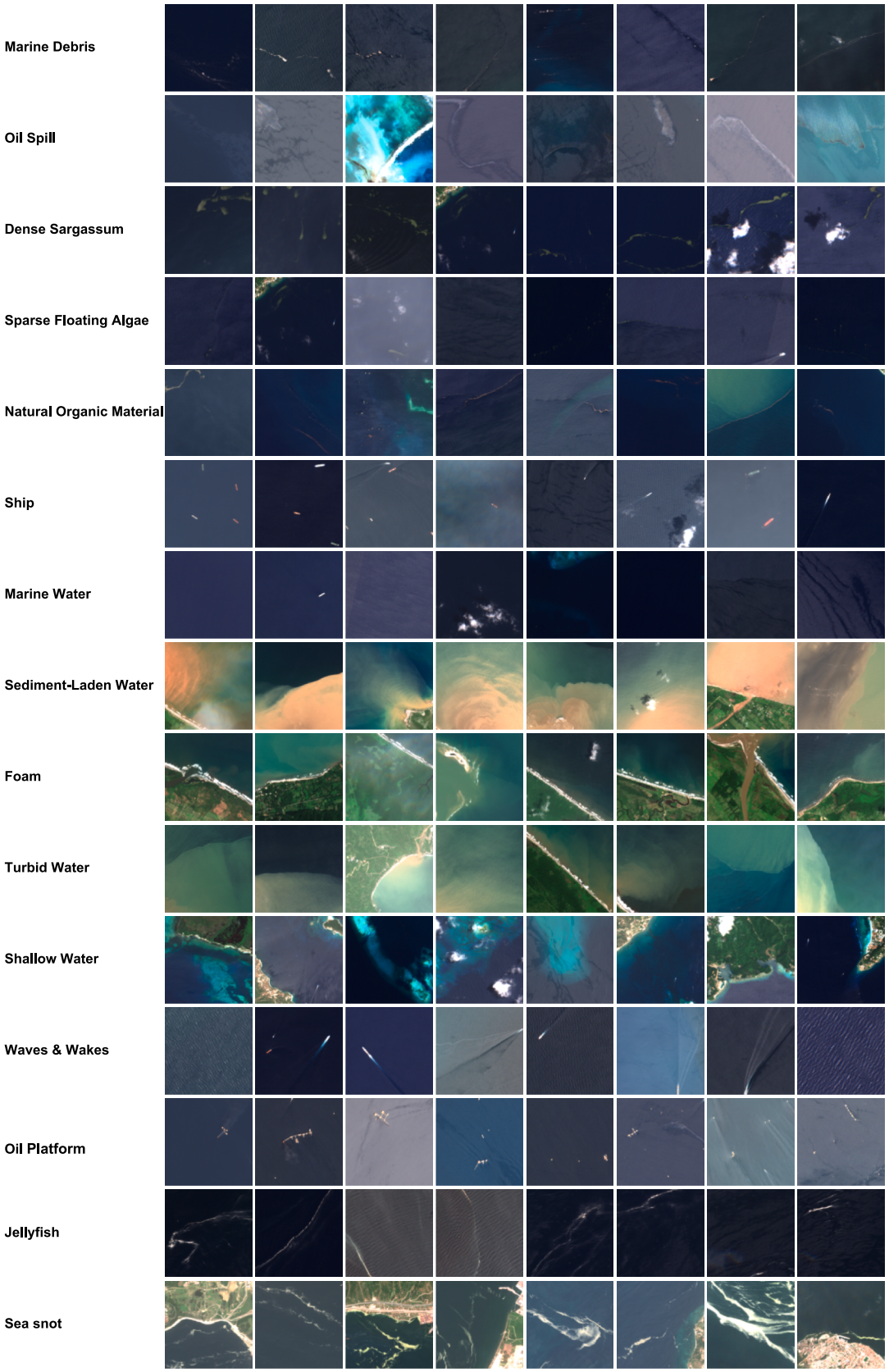


Fig. 2. An overview of MADOS provided patches. Marine pollutants and Sea surface features were annotated under various weather and sea state conditions.

do not represent spectral endmembers.

From this point in the paper, we will use the acronyms presented in Table 3. The Table defines these acronyms to facilitate clarity throughout the text. Overall, MADOS consists of 1,481,155 annotated pixels composed of 4696 *MD* pixels and 234,568 *Oil* pixels, corresponding to 469600 m^2 and 23456800 m^2 areas, respectively. Additionally, 2803 image patches are offered, each one covering 2.4x2.4 km^2. In Table 3, we observe that in MADOS, the distribution of samples per class at the pixel- or patch-level is not uniform. As expected, water-related classes can be found and digitized to a significantly larger extent than the usually small floating objects like floating plastics. Specifically, the class *MWater* has been annotated in 1360 patches, while the second-highest number of 472 patches was *MD* and the third-highest number of 361 patches was *Oil*. The considerable amount of annotated images indicates the high diversity of annotated *Oil* and *MD* pollutants. On the other hand, *Jellyfish* and *NatM* are the least annotated classes. For these classes, a limited number of reports containing sufficiently large quantities capable of being identified in S2 satellite images were collected.

3.2. Machine learning classification frameworks

In subsection 3.2.1, we present and discuss well-established and state-of-the-art ML models employed as baselines in our experiments. Subsection 3.2.2 then describes in detail our MariNeXT framework. Finally, we describe our experimental design, including the evaluation on MADOS and implementation details in subsection 3.2.3.

3.2.1. Baselines

Random Forest: Random Forest (Breiman, 2001) is a traditional shallow learning algorithm that has been frequently used for several classification tasks by the remote sensing community. A RF classification algorithm includes many decision trees and uses averaging to improve the predictive performance. We examined two different baseline models using RF: i) one based on just the spectral signatures, and ii) one with spectral signatures, spectral indices and other spatial features. In particular, the employed spectral indices were: NDVI, NDWI, FAI, FDI, Shadow Index (SI), Normalized Difference Moisture Index (NDMI), Bare Soil Index (BSI) and NRD. Following Mikeli et al. (2022), for the spatial features, we utilized the grayscale images derived by RGB composites and computed: the GrayLevel Co-occurrence Matrix (GLCM), the Gaussian at multiple scales, the Sobel of the Gaussians, and the Eigenvalues of the Hessian Matrix of the Gaussians.

U-Net: We employed U-Net as a DL baseline, which is a popular encoder-decoder convolutional architecture developed for the task of semantic segmentation (Ronneberger et al., 2015). In particular, the encoder path consists of successive blocks representing the input image. Each encoder block contains 3 x 3 convolutions for feature extraction and max-pooling layers for down-sampling. At each down-sampling step, the number of feature channels is doubled while the spatial dimensions are reduced by half. The decoder takes the representation and generates the semantic segmentation output using successive up-convolution layers.

SegNeXt: We also employed a state-of-the-art encoder-decoder architecture named SegNeXt (Guo et al., 2022). The encoder consists of a sequential four-stage pyramid structure named Multi-Scale Convolutional Attention Network (MSCAN). Each stage contains a down-sampling step followed by multiple blocks. The multiple blocks are inspired by a ViT-like (Vision Transformer) structure (Dosovitskiy et al., 2020), with the main difference being that the Multi-Scale Convolutional Attention (MSCA) replaces the self-attention module. The MSCA module consists of a depth-wise 5 x 5 convolution layer followed by a multi-branch depth-wise strip convolution layer that efficiently approximates the 7 x 7, 11 x 11 and 21 x 21 large-kernel convolutions, extracting information at multiple scales. The down-sampling steps consist of 3 x 3 convolution and batch normalization layers. The spatial dimensions of the first stage features are H/4 x W/4 and are further reduced by half for each following stage, where W and H are the width and height of the input image, respectively. Moreover, the decoder concatenates the output feature maps extracted from the last three stages, discarding the features from the first stage. Guo et al. (2022) highlight that stage-one features contain too much low-level information that does not perform well in their evaluation setup with RGB images of everyday objects, humans and actions. In order to capture global context, the Hamburger (Geng et al., 2022) module is adopted, which factorizes the input features using low-rank matrix decomposition, discarding the irregular noise and redundant features while keeping the clean signal subspace.

3.2.2. MariNeXt

A semantic segmentation framework was designed based on recent convolutional attention mechanisms (Guo et al., 2022) that are able to encode contextual information effectively depicted in multispectral

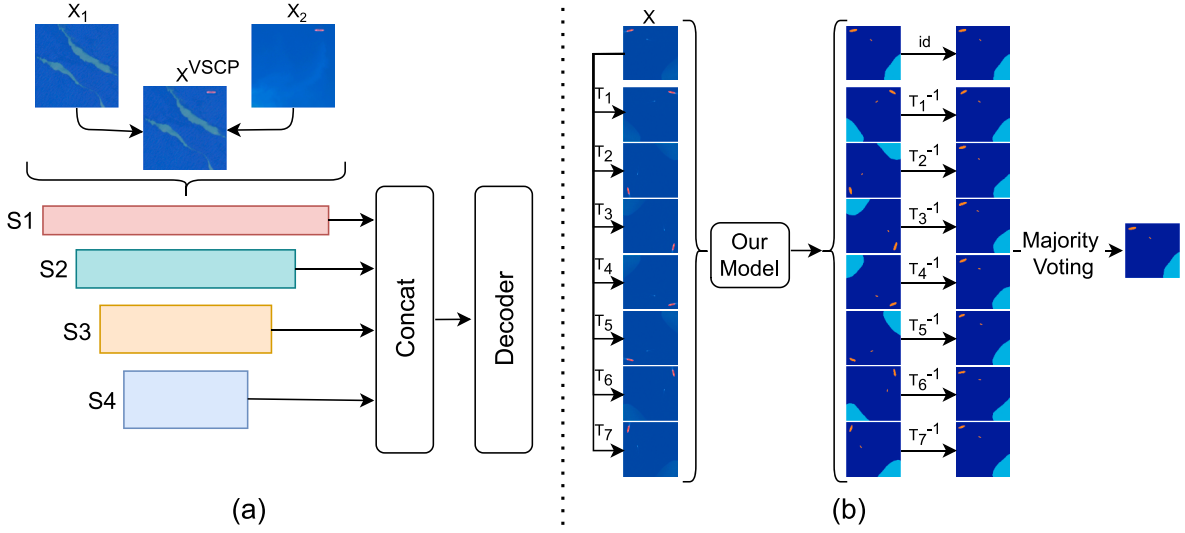


Fig. 3. Simplified overview of MariNeXt, which consists of two main parts: a) During the training process, the input batch of images is augmented using the VSCP module combing the samples, which are then passed as new input. Then MariNeXt exploits the S1 (1st-Stage Higher Resolution Features) to produce fine-grained predictions. b) During testing, the Test-Time Augmentation (TTA) strategy is employed to produce several augmented views of the input image and aggregate the predictions by major voting, producing refined predictions.

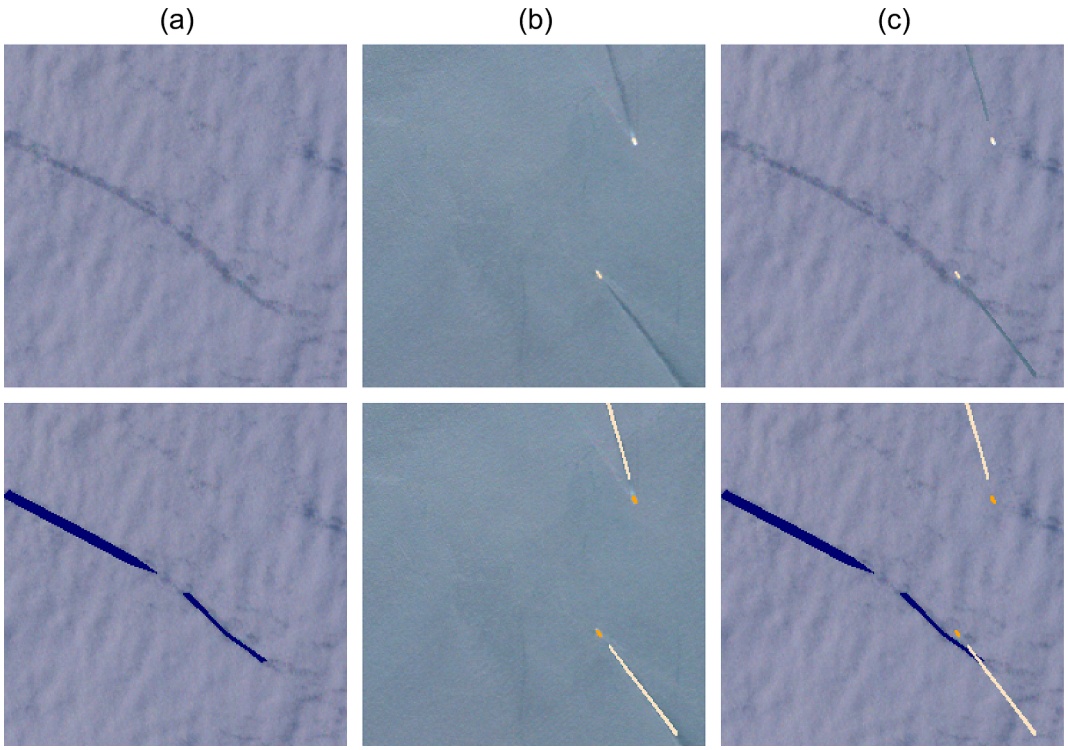


Fig. 4. MariNeXt integrates the Very Simple Copy-Paste (VSCP) augmentation technique for constructing new images, enriching MADOS. Given pairs of input images and annotations, VSCP combines the input and annotated pixels of two images to create a new synthetic input pair increasing the class co-occurrence of input data. The two input images show: a) Sea state look-alikes and Marine Water and b) Ships, Wakes and Marine Water. We also demonstrate the c) synthesized image. RGB images (top) and our corresponding annotations (down) are presented.

satellite imagery in the marine environment. In particular, we have adopted the aforementioned SegNeXt architecture and proposed improvements in three directions, i.e., modeling, training and testing (Fig. 3). These improvements were tailored for marine mapping applications and significantly boosted the performance of marine pollution detection. The three directions are described in the following paragraphs.

1st-Stage Higher Resolution Features: Regarding the modeling part, we reexamine the design of SegNeXt that discards the first-stage features. This implementation choice leads SegNeXt to output H/8 x W/8 prediction maps, which are nearest-neighbor upsampled to match the input H x W image. As shown by Guo et al. (2022), the high-resolution features might contain too much low-level information for the RGB images of everyday objects. However, in our case, various floating materials such as *Marine Debris* might be presented only on a few pixels turning high-resolution features essential for effective modeling. Additionally, this approach enables the model to output H/4 x W/4 prediction maps. Thus, we show that exploiting additional high-resolution features from the encoder improves the performance in the considered marine environment mapping task.

Very Simple Copy-Paste (VSCP): As discussed in subsections 2.2 and 3.1.2, the collection of ground truth data is challenging, involves high-cost laborious work, while collecting high-quality and confident annotations is not straightforward. In such situations where training data are scarce, incorporating data augmentation techniques can guide the model to learn better feature representations. Towards this direction, we have integrated a robust data augmentation strategy that fits perfectly the particularities of marine classes.

More specifically, Copy-Paste augmentation approaches are established techniques for object detection and instance segmentation tasks (Dvornik et al., 2018; Dwibedi et al., 2017; Fang et al., 2019). These methods are related to MixUp (Zhang et al., 2022) and CutMix (Yun et al., 2019) augmentation strategies. However, compared to MixUp and CutMix, Copy-Paste methods do not transfer an object's whole bounding box but only the exact annotated pixels. Furthermore, as everyday objects depicted in RGB images are placed in specific and oriented positions, most previous works on Copy-Paste utilized the surrounding context to place the copied object in the appropriate environment. Recently, Simple Copy-Paste (Ghiasi et al., 2021) proposed placing the objects in random positions and demonstrated promising results in instance segmentation.

In our case, MADOS consists of S2 satellite images captured at 786 km mean orbital altitude in the aquatic environment. This means that the surrounding context and the relative scale between the objects are always the same. Note also that every floating material can co-occur with any other over the same background environment (i.e., *Water*). For these reasons, in our case, there are not signifi-cantly different surrounding contexts or scale issues. Therefore, inspired by Ghiasi et al. (2021), we propose a straightforward and Very Simple Copy-Paste (VSCP) training augmentation approach that fits excellently in our *semantic segmentation* task using sparse annotations (Fig. 4). In particular, given a batch $X \in \mathbb{R}^{(N_{batch},C,W,H)}$ and $Y \in \mathbb{Z}^{(N_{batch},W,H)}$, where X are the input images, Y the groundtruth masks, N_{batch} is the batch size, C is the spectral dimension, and W and H are the width and height, respectively, our approach augments the current training batch size by $N_{VSCP} = \lfloor \frac{N_{batch}}{2} \rfloor$, where $\lfloor \bullet \rfloor$ outputs the greatest integer less than or equal to the input. Specifically, given the i and the $i+N_{VSCP}$ samples in the current training batch, we construct a new augmented image by adding the annotated regions and replacing the pixels to the image i from the image $i + N_{VSCP}$. The new synthesized image contains the annotations from both i and $i+N_{VSCP}$ using as background the i image (Fig. 3). More formally, each pixel element $x^{VSCP}_{n,c,w,h}$ *and* $y^{VSCP}_{n,w,h}$ of the N_{VSCP} augmented images $X^{VSCP} \in \mathbb{R}^{(N_{VSCP},C,W,H)}$ with the corresponding ground truth masks $Y^{VSCP} \in \mathbb{Z}^{(N_{VSCP},W,H)}$, are:

$$x^{VSCP}_{n,c,w,h} = \begin{cases} x_{n+N_{VSCP},c,w,h} & \text{if } y_{n+N_{VSCP},w,h} \neq -1 \\ x_{n,c,w,h} & \text{otherwise} \end{cases}, \quad (1)$$

$$y^{VSCP}_{n,w,h} = \begin{cases} y_{n+N_{VSCP},w,h} & \text{if } y_{n+N_{VSCP},w,h} \neq -1 \\ y_{n,w,h} & \text{otherwise} \end{cases}, \quad (2)$$

for $n = 1, \cdots, N_{VSCP}$, $c = 1, \cdots, C$, $w = 1, \cdots, W$ and $h = 1, \cdots, H$. Note that the non-annotated pixels have $y_{n,w,h} = -1$.

Test-Time Augmentation (TTA): Similarly to Gonzalo-Martín et al. (2021); Moshkov et al. (2020); Wang et al., (2019b,2019a), during the training procedure, we have also considered augmentations during the inference (test-time) execution in order to refine MariNeXt predictions by averaging the outputs across the different augmented views. To do so, we have explored a simple and effective Test-Time Augmentation (TTA) strategy.

In particular, for each input image $X \in \mathbb{R}^{C\times W\times H}$, a trained model $f : \mathbb{R}^{C\times W\times H} \rightarrow \mathbb{N}^{W\times H}$ produces an output $Y \in N^{W\times H}$ forming the assigned integers that correspond to the predicted class for each pixel. Using N_T different transformations $\{\mathscr{T}_i\}_{i=1}^{N_T}$ we obtain a set of predictions $\{Y^{\mathscr{T}_i}\}_{i=1}^{N_T}$:

$$Y^{\mathscr{T}_i} = \mathscr{T}_i{}^{-1} f(\mathscr{T}_i(X)), \text{for } i = 1, \cdots, N_T \quad (3)$$

Then for each pixel we aggregate the results using the majority voting which corresponds to the mode of the sampling distribution:

$$y^{TTA}_{w,h} = MODE\left(\left\{Y^{\mathscr{T}_i}_{w,h}\right\}_{i=1}^{N_T}\right), \text{for } w = 1, \cdots, W \text{ and } h = 1, \cdots, H \quad (4)$$

The final predicted image is the $Y^{TTA} \in N^{W\times H}$ with elements $y^{TTA}_{w,h}$. In order to keep the initial spectral bands and object scales as in the raw S2 images, we do not apply any color or scale transformation. Specifically, we combine rotations of 0°, 90°, 180°, 270° with the horizontal flipping, resulting in $N_T = 8$ different transformations applied in test-time execution.

3.2.3. Experimental design

Evaluation on MADOS. In order to train and assess MariNeXt and the other baseline models, we split MADOS into three disjoint train, validation and test sets. The train set is employed to train the models explicitly, the validation set to monitor the learning process and the test set to perform our evaluation. Each of these sets retains the class distribution of the whole MADOS dataset. Additionally, the cropped patches from each scene/ unique date are kept on the same set. The splits contain 1433 training (761887 pixels), 642 validation (358121 pixels) and 728 test (361147 pixels) patches. The number of patches and pixels corresponds to approximately 50 %/25 %/25 % of the initial MADOS dataset. We note that each data split was created as a representative subset of the entire MADOS dataset in terms of pixels and patches statistics (Table S2).

To assess the models' performance quantitatively, we relied on three evaluation metrics: i) Jaccard Index or Intersection-over-Union (IoU), ii) F_1 score, iii) Overall Accuracy (OA), and iv) per class Recall. The evaluation metrics were calculated as follows:

Table 4
Evaluation of MariNeXt against ML baselines on MADOS test set using F_1, mean Intersection of Union (mIoU) and Overall Accuracy (OA) metrics, averaged over five trials. The highest scores are highlighted with bold fonts.

Method	F_1 (%)	mIoU (%)	OA (%)
RF	56.6	43.9	67.1
RF^{++} (Mikeli et al., 2022)	64.4	52.4	83.8
U-Net (Ronneberger et al., 2015)	63.8	51.0	82.9
SegNext (Guo et al., 2022)	60.6	49.2	86.6
MariNeXt	**76.0**	**64.3**	**89.1**

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \quad (5)$$

where TP_c are true positives, FP_c are false positives and FN_c are false negatives of each c class.

Since we deal with a multi-class task, the evaluation metric for the overall assessment is the mIoU which is the average IoU over all classes c:

$$mIoU = \frac{1}{n}\sum_c^n IoU_c, \quad (6)$$

The second metric is the average for each class F_1 score (Macro-F_1/ mF_1), which is the harmonic mean between Precision $P = TP/(TP + FP)$ and Recall $R = TP/(TP + FN)$:

$$mF_1 = \frac{1}{n}\sum_c^n F_{1c} = \frac{1}{n}\sum_c^n \frac{2P_cR_c}{P_c + R_c}. \quad (7)$$

We also considered the Overall Accuracy:

$$OA = \frac{\sum_c^n TP_c}{\sum_c^n TP_c + FP_c}. \quad (8)$$

Implementation details. In all experiments, we upsampled the 20 m and 60 m bands to 10 m using the nearest-neighbor resampling approach. For both RF models, we followed the training setup as proposed by Kikaki et al. (2022); Mikeli et al. (2022). More specifically, we employed 125 trees, each with a maximum depth of 20 nodes and used a class weighting scheme to handle the class imbalance while the annotators' confidence levels were also exploited. Regarding the spatial features, the grayscale images derived by RGB composites were utilized in order to extract the GLCM features using a 13 x 13 sliding window. We computed the Gaussian, the Sobel of the Gaussians, and the Eigenvalues of the Hessian Matrix of the Gaussians at the different scale levels using the standard deviation of $\sigma = 1, 2, 4, 8, 16$.

For the DL models, we based our implementation on MARIDA (Kikaki et al., 2022), SegNext (Guo et al., 2022), timm (Wightman, 2019) and mmsegmentation (MMSegmentation Contributors, 2020) libraries. For all models, we adapted the first input layer to handle the 11 S2 bands and the final layer to output the MADOS 15 classes. Regarding SegNeXt, we used the *Tiny* (SegNeXt-T) version. The same was adopted in MariNeXt. All DL models were trained from scratch with Adam algorithm for 80 epochs with an initial learning rate of $2x10^{-4}$ minimizing the Cross-Entropy loss. After epochs 45 and 65 the learning rate was set to $2x10^{-5}$ and $2x10^{-6}$, respectively. As detailed in Table S3, we set the batch size for U-Net to 15 and for SegNeXt and MariNeXt to 5. For data augmentations, we used 0°, 90°, 180° or 270° random rotations as well as horizontal and vertical flips. Following MARIDA, we used the same weighting scheme to alleviate the effects of the imbalanced class distribution. During training, we maintained the *exponential moving average* (EMA) of the trained parameters θ by according to $\theta' \leftarrow \alpha\theta' + (1-\alpha)\theta$, setting $\alpha = 0.999$. We found this process extremely beneficial to produce smooth loss curves as well as robust and stable models.

4. Experimental results and validation

In this section, we compare our method with the other employed ML models and evaluate its performance both quantitively and qualitatively. In addition, an ablation study is performed to assess the main components of the proposed MariNeXt framework.

4.1. Comparing MariNeXt with ML baselines

Firstly, the performance of all applied models is quantitatively evaluated. Overall, we observe that MariNeXt achieved the best classification performance, as indicated by the highest average scores for all

Table 5
The resulting Confusion Matrix after applying MariNeXt on MADOS test set. The metrics of Recall, IoU, Precision and F_1 are presented. Columns represent the predicted labels and rows represent the reference labels (%). All acronyms are stated in Table 3.

Ground Truth \Predicted	MD	DenS	SpFA	NatM	Ship	Oil	MWater	SLWater	Foam	TWater	SWater	Waves	OilPlat	Jelly fish	Sea snot	# of pixels (test set)	Recall (%)
MD	**70.4**	0.0	2.1	0.0	11.1	0.1	12.7	0.0	0.0	0.0	0.0	3.1	0.4	0.0	0.0	973	**70.4**
DenS	0.5	**70.0**	28.2	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1303	**70.0**
SpFA	1.2	1.3	**90.4**	0.2	0.0	0.0	6.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1786	**90.4**
NatM	22.7	0.0	5.3	46.1	17.6	0.1	5.7	0.0	0.0	0.2	0.0	2.3	0.0	0.0	0.0	278	46.1
Ship	3.1	0.0	0.0	0.0	**52.7**	2.1	0.3	0.2	0.0	0.4	0.0	1.6	39.3	0.3	0.0	3574	**52.7**
Oil	0.0	0.0	0.0	0.0	0.0	**70.6**	28.3	0.0	0.0	0.1	0.2	0.7	0.0	0.0	0.0	51,528	**70.6**
MWater	0.1	0.0	0.0	0.0	0.0	2.6	**95.7**	0.0	0.0	1.1	0.2	0.4	0.0	0.0	0.0	139,465	**95.7**
SLWater	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**99.3**	0.0	0.7	0.0	0.0	0.0	0.0	0.0	93,254	**99.3**
Foam	0.5	0.0	0.0	0.0	4.9	0.1	0.3	0.0	**80.6**	1.1	0.0	7.4	0.0	0.0	5.2	387	**80.6**
TWater	0.0	0.0	0.0	0.0	0.1	0.0	1.5	0.1	0.0	**89.8**	7.6	1.0	0.0	0.0	0.0	42,168	**89.8**
SWater	0.0	0.0	0.0	0.0	0.0	0.0	26.1	0.0	0.0	12.4	**61.2**	0.3	0.0	0.0	0.0	12,732	**61.2**
Waves	2.9	0.0	0.0	0.0	2.2	17.3	41.7	0.0	0.1	1.2	6.1	28.4	0.1	0.0	0.0	6401	28.4
OilPlat	0.1	0.0	0.0	0.0	3.5	1.1	0.3	0.0	0.0	0.0	0.0	0.5	**94.6**	0.0	0.0	4790	**94.6**
Jellyfish	1.1	0.0	0.0	0.0	0.0	1.5	0.0	0.0	0.0	0.0	0.0	13.1	0.0	**84.4**	0.0	274	**84.4**
Sea snot	22.4	0.0	0.0	0.0	0.0	7.0	0.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	**69.6**	2234	**69.6**
IoU (%)	34.6	68.6	73.0	42.7	45.9	64.4	82.9	99.2	77.3	82.1	47.1	23.2	72.9	81.8	68.9	**mIoU:**	**64.3**
Precision (%)	40.6	97.7	79.2	83.8	77.9	88.8	86.2	99.9	95.0	90.8	67.9	57.1	76.1	96.2	98.7	**OA:**	**89.1**
F1-score (%)	51.3	81.2	84.3	57.2	62.8	78.3	90.7	99.6	87.1	90.1	63.7	37.7	84.3	89.9	81.4	**F1-macro:**	**76.0**

evaluation metrics, i.e., 76.0 % for F_1, 64.3 % for mIoU and 89.1 % for OA (Table 4). MariNeXt outperformed all baselines by at least 11.6 % in F_1 and 11.9 % in mIoU metrics.

The second highest scores were obtained by RF^{++} (i.e., 64.4 % for F_1 and 52.4 % for mIoU), followed by U-Net, which led to 63.8 % for F_1 and 51.0 % for mIoU. Regarding the rest two classifiers, the default RF reached the lowest scores for all metrics while SegNeXt achieved an improvement of + 4.1 % for F_1, +5.3 % for mIoU and + 19.5 % for OA. We note that the default SegNeXt achieved the second-highest OA score (i.e., 86.6 %) (Table 4).

4.2. Quantitative and qualitative evaluation per thematic class

To further assess the performance of the MariNeXt and study the agreement between the annotated data and the predictions, we present and discuss the corresponding confusion matrix (Table 5). The scores are demonstrated for all metrics per class on the test set. Our analysis and discussion focus on the achieved F_1 scores, but similar conclusions can also be derived from the other metrics. Specifically, the highest scores (i.e., >80 %) for F_1 were achieved for classes with typical optical characteristics and spectra, such as *SLWater*, *DenS*, *SpFA*, water-related classes (i.e., *MWater*, *TWater*), classes with distinct textural patterns such as *OilPlat* and *Foam*, as well as floating organisms like *Jellyfish* and *Sea snot*.

On the other hand, *Waves*, *MD* and *NatM* showed the least F_1 scores (i.e., <58 %). For *Oil*, *Ship* and *SWater*, our method led to F_1 values > 62 %.

More specifically, one can observe that *MD*, a quite challenging case, presented relatively low rates for F_1 (i.e., 51.3 %) and IoU (i.e., 34.6 %). Nevertheless, 70.4 % of its testing pixels were correctly classified, indicating that there is a high possibility that our method detects *MD* accurately. Also, 11.1 % were classified as *Ship* and 12.7 % as *MWater*. This can be attributed to the fact that plastics are spectrally similar to ships, as well as they can be probably spectrally close to *MWater* under certain circumstances (e.g., low density, submersion, polymer type, environmental conditions).

The *Oil* class, consisting of several oil types, presented 78.3 % F_1 and 64.4 % IoU. In addition, 70.6 % of its testing data were correctly predicted as *Oil*. However, a proportion of 28.3 % for *Oil* testing labeled pixels were classified as *MWater*, indicating similar spectral behavior.

Regarding artificial materials and man-made objects, *OilPlat,* characterized by distinct spatial structures, achieved a high F_1 rate (i.e., 84.3 %) and a very high Recall value (i.e., 94.6 %). On the contrary, *Ship* class composed of varying sizes and cruising speeds reached a 62.8 % F_1 score. Almost half of the testing annotated data were correctly classified as *Ship*, while 39.3 % were wrongly predicted as *OilPlat*. Apparently, these two classes share common spectral characteristics due to the similar materials they consist of; thus, their discrimination is considered challenging.

Water-related classes, such as *SLWater*, *MWater* and *TWater* held high F_1 scores of over 90 %. *Foam* also achieved high F_1 score of over 80 % due to its distinct spatial pattern that presents in coastal areas (wave breaking). The least accurate water-classes were *SWater* and *Waves*, with 63.7 % and 37.7 % F_1 scores, respectively. Recall values were 61.2 % for *SWater* and 28.4 % for *Waves*. This was mainly affected by the confusions between *SWater* with *MWater* or *TWater*; and *Waves* with *MWater* or *Oil*.

Floating macroalgae species presented high F_1 rates over 80 %. However, as Table 5 indicates, there is some misclassification cases between *DenS* and *SpFA*. Specifically, 28.2 % of *DenS* testing annotated data were predicted as *SpFA*. In fact, this was expected since both classes belong to the same super-class of floating macroalgae which contains mainly *Sargassum*.

Despite the number of testing pixels being the lowest compared to the other categories, *Jellyfish* achieved a F_1 score of almost 90 %. As Table 5 shows, 13.1 % of the testing data were classified as *Waves,* a class

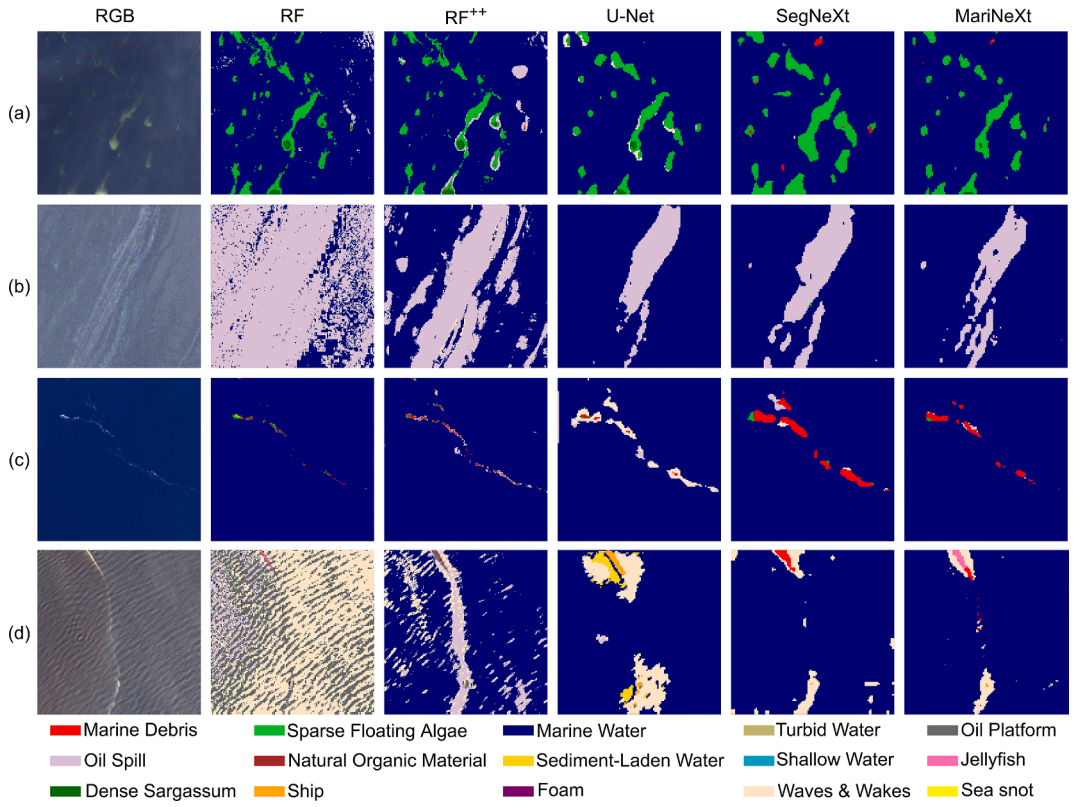


Fig. 5. Prediction maps extracted by the applied baselines and MariNeXt. The RGB images show: a) Dense & sparse Sargassum, Marine Water and haze/thin clouds, b) Oil Spill and Marine Water, c) Marine Debris and Marine Water, and d) Jellyfish and rough sea state. The RGB images were stretched for illustration purposes. For more classification results, the reader is referred to Supplementary Figure S1.

Table 6
Ablation study on the proposed approach. SegNeXt is considered as a baseline for the ablation.

1st-Stage Features	VSCP	Increased BS	TTA	F_1	mIoU
×	×	×	×	60.6	49.2
×	×	×	✓	63.4	51.6
×	✓	×	×	65.9	53.0
✓	×	×	×	66.3	53.6
✓	×	✓	×	66.1	53.6
✓	×	×	✓	67.3	55.6
×	✓	×	✓	69.0	56.2
✓	✓	×	×	72.7	59.6
✓	✓	×	✓	**76.0***	**64.3***

VSCP: Very Simple Copy-Paste, BS: Batch Size, TTA: Test-Time Augmentation, *MariNeXt.

of relatively high-intensity values in the optical spectrum. The *Sea snot* class also reached a high F_1 value of 81.4 %, while 22.4 % of its testing labeled data were classified as *MD*. Moreover, the quite challenging *NatM* class, which contains vegetation and wood materials, resulted in a F_1 score of 57.2 %. Again, we note that 22.4 % of its testing annotations were classified as *MD*.

Apart from the quantitative assessment, a qualitative evaluation of the applied models on the test set was also performed. To describe the predictive power of MariNeXt and employed baselines, we demonstrate representative classification results in Fig. 5. By inspecting all resulting maps, it is clear that MariNeXt achieved the highest predictive performance.

In particular, it led to high-resolution results, being the most robust to isolated pixels and weather conditions (Fig. 5). Especially in the case of *Oil*, our model could accurately discriminate *Oil* from *MWater* in fine detail (Fig. 5b). Although MariNeXt resulted in coarser outputs than RF models due to its convolutional nature, it also managed to predict small objects such as *Jellyfish* (Fig. 5d).

We generally observe an agreement between our MariNeXt prediction results and metric scores. For instance, a small part of *Sargassum* pixels of low biomass densities was misclassified as *MD* (Fig. 5a), some *MWater* pixels around debris were predicted as *Oil* (Fig. 5c), and sparse *Jellyfish* pixels were classified as *MD* (Fig. 5d). These misclassification cases were also observed in the quantitative results in Table 5.

Regarding the other ML baselines, it is observed that SegNeXt led to similar classification results with MariNeXt but with relatively coarser outputs. This also supports our framework design choices by including first-stage features and exploiting their high-resolution feature representations. U-Net also yielded satisfactory results for *Sargassum* and *Oil* while it misclassified several *MD* pixels as *Waves* and almost all *Jellyfish* as *Ship*. Concerning RF models, the default RF, which reached the lowest scores for all metrics (Table 4), failed to discriminate *Oil* from *MWater* and *Waves* from *MWater*, presenting the highest sensitivity to S2 sensor noise and isolated pixels (Fig. 5b,d). However, adding spectral (indices) and texture/ spatial information substantially improved the classification performance results. In fact, RF^{++} reached the second highest metric scores (Table 4), providing high-resolution outputs for *Sargassum* and *Oil* classes (Fig. 5a,b). It also achieved to predict some pixels correctly as *Waves*. On the other hand, in the case of small-width objects (i.e., *MD* and *Jellyfish*), it led to misclassification results as part of *MD* pixels was identified as *SpFA*, *Waves* and *Oil,* and *Jellyfish* was classified as *OilPlat*, *MD* and *Ship* (Fig. 5c,d).

By evaluating the baselines qualitatively and quantitatively, we observe that although RF^{++} led to the second-highest rates (Table 4), its extracted maps presented enough scattered, isolated predictions. This points out that RF^{++} is prone to artifacts (e.g., S2 mixed bands, low-density pixels) and weather conditions. Besides, the qualitative interpretation showed that SegNeXt could capture the semantic context of

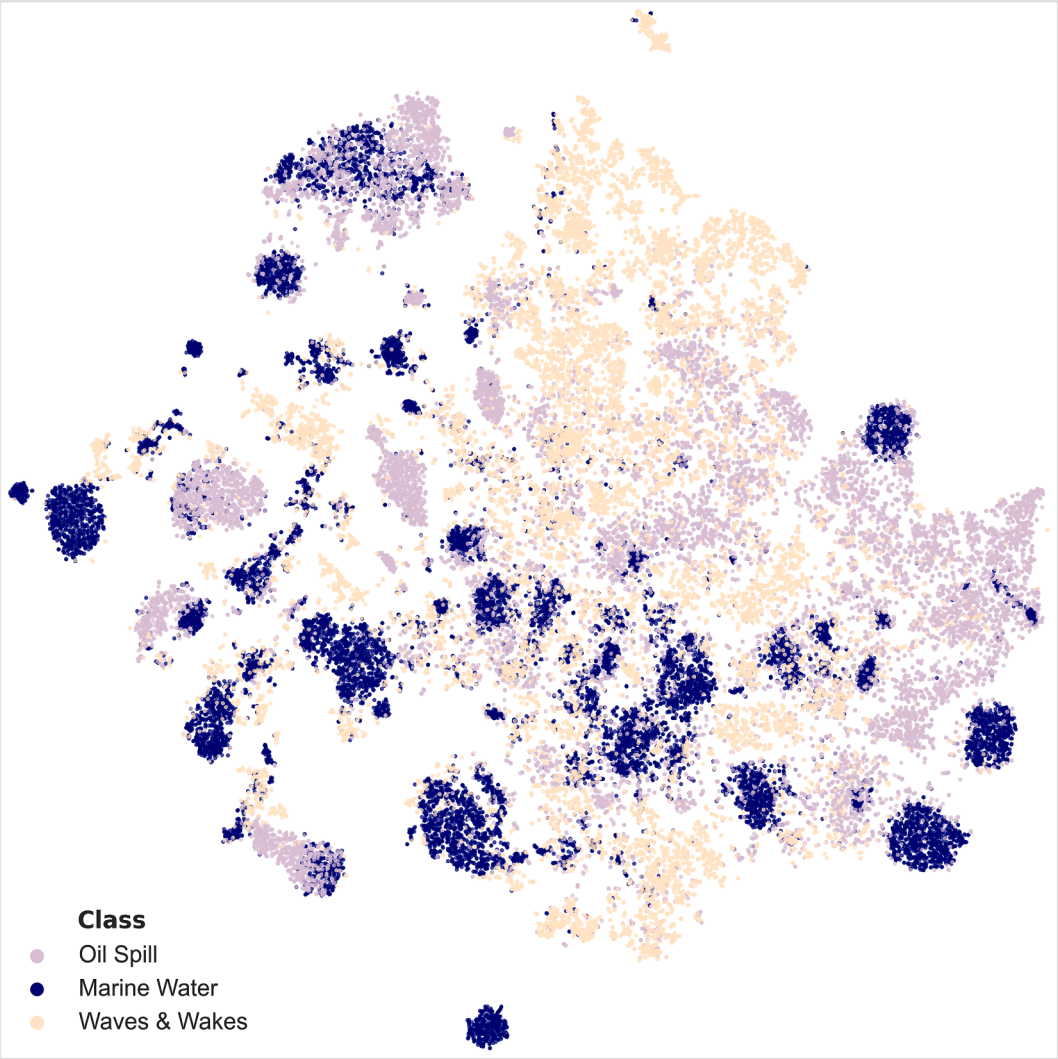


Fig. 6. T-SNE Analysis on MADOS dataset using Spectral Angle Mapping for Oil Spill, Marine Water and Waves & Wakes.

depicted classes. Note that SegNext produces H/8 x W/8 prediction maps, which are simply nearest-neighbor upsampled. Therefore, it produces coarse-grained prediction maps leading to relatively low F_1 and mIoU scores (Table 4).

4.3. Ablation study for MariNeXt

In order to demonstrate the impact of the design choices regarding the proposed MariNeXt framework, we performed a detailed ablation by adding/ removing different components. We report the corresponding F_1 and mIoU values for each experiment in Table 6. After a close look at all experiments, we observe that TTA is an essential feature, as it improved the model's performance in all experiments, i.e., either adding 1st-Stage Features or not, either with VSCP or not. In particular, by integrating this augmentation strategy, the model produced more robust predictions based on eight different augmentations of the same image. Without adding features of the 1st stage, TTA resulted in an increase of 2.8 % for F_1 and 2.4 % for mIoU, as with the additional features the corresponding values were 1 % and 2 %, respectively.

Additionally, our VSCP augmentation strategy was critical for better classification results (Table 6). This was expected since increasing the training samples improves the model's accuracy and generalization. In the case of default SegNeXt, an improvement of 5.3 % for F_1 and 3.8 % for mIoU was obtained, as for the extra high-resolution features, an enhancement of 6.4 % for F_1 and 6 % for mIoU was performed. Table 6 also shows the importance of utilizing higher-resolution features of the 1st stage. The additional features led to an improvement of 5.7 % for F_1 and 4.4 % for mIoU compared to the default model.

We further investigated whether the enhanced performance with the additional mixed samples was due to the increased batch size. Thus, increasing the batch size with unmixed samples was also examined, leading to lower metric scores (Table 6).

As evident in Table 6, the best combination is 1st-Stage Features, with VSCP and TTA, which shows the effectiveness of our method over the baseline SegNeXt. Also, we highlight that each component is orthogonal to the others, as each one benefits the performance independently. Last but not least, we experimented with four different BSs (5, 10, 15, 20), indicating that the batch size selection was crucial for the performance of the applied models (Table S3).

To further understand the effectiveness of MariNeXt's components, we systematically enabled/ disabled them from our framework inspecting the class-wise F_1 scores (Table S4). By integrating high-resolution 1st stage features, F_1 scores were significantly increased by over 12 % for the *MD*, *DenS*, *SpFA* and *Foam* classes. High-resolution features were beneficial for our task as we focused on pixel-based predictions with objects presented at significantly different scales (e.g., *Oil* vs. *MD*) Furthermore, by enabling TTA and VSCP, we observe an increase of over 20 % for *NatM*, *Jellyfish* and *Foam* (Table S4), which are

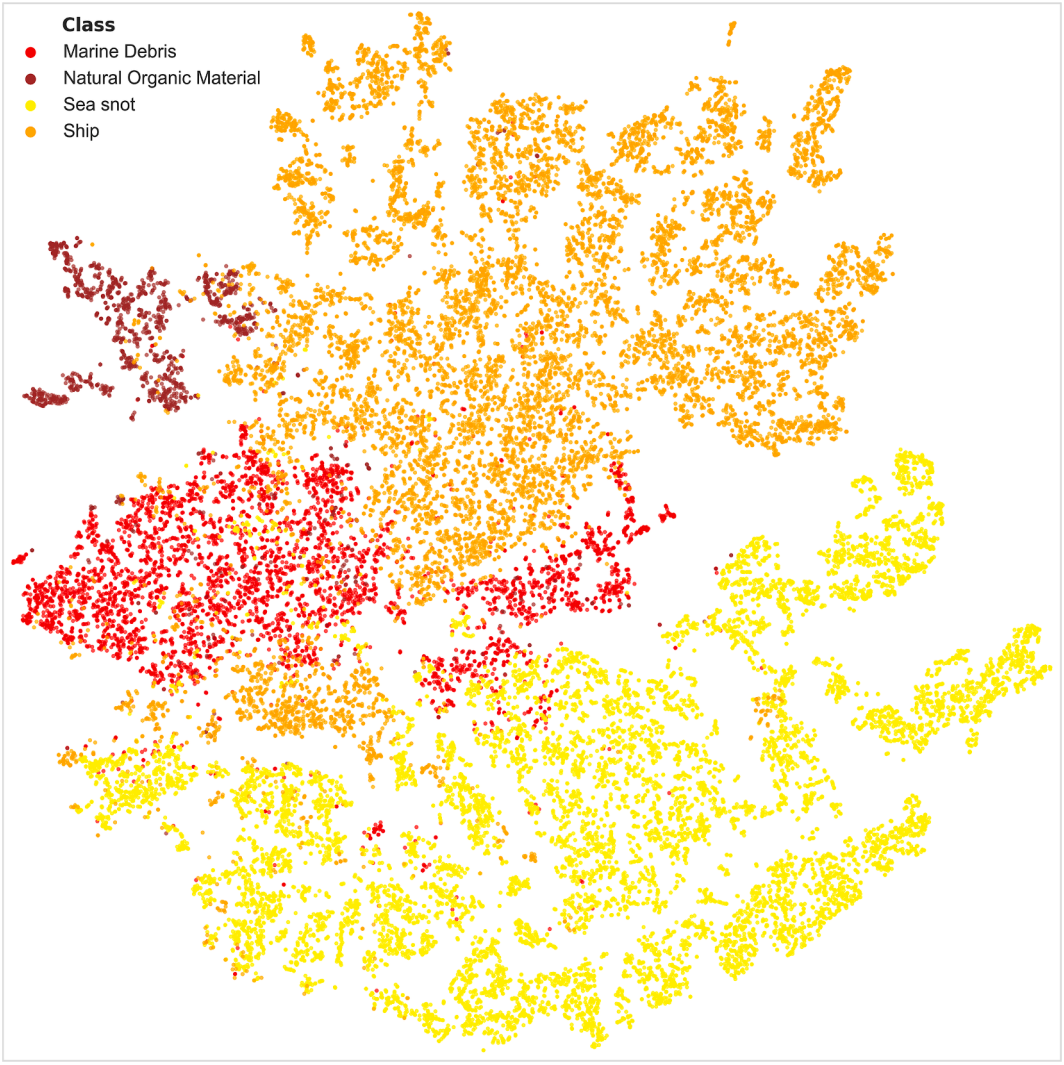


Fig. 7. T-SNE Analysis on MADOS dataset using Spectral Angle Mapping for Marine Debris, Ship, Natural Organic Material and Sea snot.

the most under-representative classes in the MADOS dataset. By integrating TTA, MariNeXt delivers more robust predictions, improving the classification performance. At the same time, with VSCP, MariNeXt synthesizes additional images on the fly, tackles unbalanced data and strengthens its performance on our *semantic segmentation* task.

5. Discussion

5.1. Potentials in detecting and discriminating marine pollutants in S2 data

Based on our experimental results, the MariNeXt framework can distinguish the two primary marine pollution forms, i.e., *MD* and *Oil*, on optical S2 images. The quantitative assessment presented in Table 5 indicated that only 0.1 % of *MD* was classified as *Oil*, while 0.02 % of *Oil* was predicted as *MD*. Furthermore, the calculated T-SNE plot (Figure S2) indicates a clear separation potential in the 2D space between these two pollutants. Based on these experiments, we conclude that it is feasible to create effective marine monitoring solutions that can consider both classes without significant misclassification issues.

Oil Spill. Based on both quantitative (Table 5) and qualitative results (Fig. 5), we reveal the most challenging cases for *Oil*. Most of these cases correspond to sparse *Oil* regions without distinct spatial patterns classified as *MWater* (Fig. 5). Moreover, *Waves* are frequently misclassified as *Oil* (Fig. S1a), (Table 5). The main source of these misclassifications is that *Oil*, *MWater* (including sea state look-alikes) and *Waves* can be spectrally similar, leading to a significant overlap in T-SNE plot (Fig. 6). We argue that these observations are due to the high spectral variations derived from different oil concentrations and weathering processes. These outcomes are in accordance with the literature since *Oil* has been proven to present spectral patterns similar to *Water* and *Waves* (Fingas and Brown, 2018; Sun et al., 2015; Trujillo-Acatitla et al., 2022). However, MariNeXt managed to tackle these similarities, achieving a relatively high F_1 of 78.3 % for *Oil*.

In order to further assess MariNeXt's efficiency, we present the learned feature space against the initial one (based on Spectral Angle Mapping) (Figure S3). Specifically, we extract the output features of the MariNeXt encoder, as they are given to the decoder, and visualize them (Figure S3b). During the training process, MariNeXt managed to learn rich representations of the input data and alleviate the challenging issue of *Oil*, *Waves* and *MWater,* by separating them to a certain extent in the feature space (Figure S3b).

Marine Debris. For *MD*, the most competing classes were *Ship*, *Sea snot* and *NatM* (Table 5, Figure S1a,b,c). As can be observed in the corresponding T-SNE plot (Fig. 7), despite the overall distinct cluster representations, there are several low-density *NatM and Sea snot pixels* that overlap with the *MD* cluster. Dense *MD* pixels are also depicted in the *Ship* cluster, which is in accordance with the literature (Kikaki et al., 2022).

Interestingly, the main *MD* and *Sea snot* clusters are distinguished (Fig. 7), suggesting that the considered classes can be differentiated under certain circumstances (e.g., Chlorophyll pigments presence), as was also reported by Papageorgiou et al. (2022). Although *MD* detection from current sensors is quite challenging (Martinez-Vicente, 2022; Politikos et al., 2023), we show that state-of-the-art deep neural networks like MariNeXt can contribute to addressing certain overlaps, e.g., *MD* against *Ship* and *Sea snot* (Figure S4). Still, the most competing feature for *MD* is *NatM* due to the usually similar spatial patterns and spectral behavior (Figure S4b).

Water-related Classes. In general, *MWater, SLWater* and *TWater* perform well with a relatively small number of false positives. However, in some cases, *SWater* is misclassified as *TWater*, and *Waves* and *SWater* are predicted as *MWater* (Table 5, Fig. 5d). The corresponding T-SNE plot (Figure S5a) also indicates that water-related classes can exhibit, in some cases, similar spectral characteristics. The most well-separated cluster was formed by *SLWater* (Figure S5a), which presents distinct spectral behavior due to the increased concentrations of suspended sediments (highest Recall in Table 5). Recent studies have also reported this outcome (Kikaki et al., 2022; Mikeli et al., 2022). On the other hand, the *Waves* class (with relatively high-intensity values) presents the greatest spectral variance derived from the different sea state conditions (Figure S5a, Fig. 6). Although MariNeXt contributes to the separation of the water-related classes, we highlight that the *Waves* class is the most challenging, and its accurate detection is crucial for the overall performance of any operational marine mapping system. In supplementary, we provide more discussion about the other classes (Text S1).

5.2. Limitations & challenges

MADOS fills a current gap in benchmark data availability for marine pollution detection, and the MariNeXt framework seems capable of classifying competing marine classes. That being said, some limitations need to be considered. First, MADOS is, by nature, an unbalanced dataset. Classes like *MWater* and *Oil* are abundant in the dataset while *Jellyfish*, *Foam* and *NatM* are less represented. As a result, MariNeXt's ability to detect these less representative classes might potentially be limited in regions beyond the dataset's coverage. MADOS can be expanded in the future with further observations from different regions to enhance the predictive capabilities of MariNeXt. This will boost the detection accuracies on these less representative classes, particularly improving the model's performance when applied to large-scale real-world applications. Furthermore, S2 data limitations (Hu, 2022), combined with diverse marine environmental conditions, and not always consistent atmospheric correction processes, led to high variability in MADOS reflectance values. These challenges need to be addressed in any operational remote sensing system. Also, during our experimental validation, we observed some systematic false positives such as *Waves* around *OilPlat*, *OilPlat* mixed with *Ship* and *MD* spikes (Figure S1). Post-processing strategies can tackle these systematic spurious correlations and keep only the main floating object.

Putting the limitations aside, MADOS is a valuable dataset that benchmarks ML algorithms for marine pollution detection from open S2 data, supporting the development of future operational marine monitoring solutions. In addition to the *semantic segmentation*, MADOS can contribute to other ML and remote sensing tasks, including semi-, self- or unsupervised and transfer learning. We also encourage the community to examine pan-sharpening methods or other upsampling techniques beyond Nearest Neighbor resampling that has been mostly employed.

6. Conclusions

This study introduces MADOS, a globally-distributed benchmark dataset for *Marine Debris* and *Oil Spill* detection from Sentinel-2 satellite imagery. MADOS consists of competing marine classes and almost 1.5 million annotated pixels, being to the best of our knowledge, the largest available Sentinel-2 dataset for marine pollutants. We have also introduced MariNeXt, a novel DL framework capable of recognizing specific spatial patterns and distinguishing competing classes that cannot be spectrally separated. Through our experiments, MariNeXt led to promising prediction maps and outperformed the other ML baselines in all quantitative metrics.

We also demonstrate that *Oil* and *MD* tend to be separable and detectable. Most false positives and negatives are related to low-density floating matters, objects composed of similar materials or presenting high-intensity values, and some water-related classes (i.e., *Waves*). By considering, without exceptions, the competing sea surface features and water-related classes, reliable operational monitoring solutions can be developed using the openly available multispectral data. Still, several challenges exist, including the scarcity of in situ data, the need for more balanced datasets, and the requirement for more sophisticated machine learning models capable of accommodating multi-temporal data to facilitate informed decision-making.

CRediT authorship contribution statement

Katerina Kikaki: Conceptualization, Data Curation, Methodology, Writing – original draft, Writing – review & editing. **Ioannis Kakogeorgiou:** Data Curation, Methodology, Software, Writing – original draft, Writing – review & editing. **Ibrahim Hoteit:** Writing – review & editing. **Konstantinos Karantzalos:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.isprsjprs.2024.02.017.

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